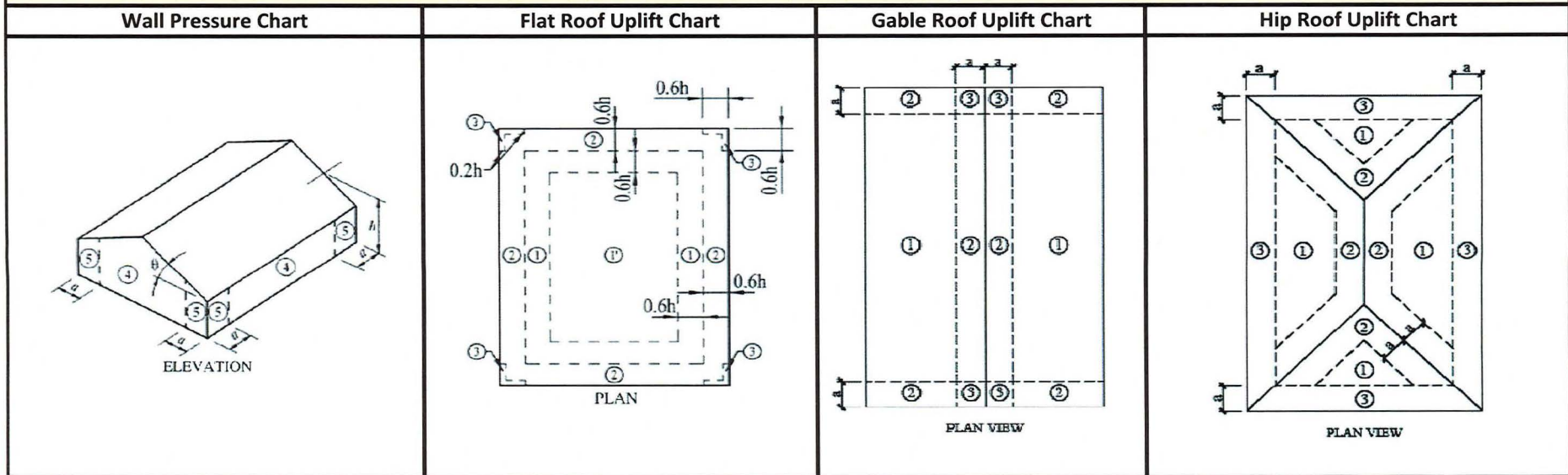


ASCE 7-22

Roof and Wall Zone Chart Diagrams



Instructions on how to use these Charts: Determine Mean Roof Height, h , which is top of roof for flat roofs or the mean roof height for pitched roofs. Find your least horizontal dimension for your building, not including a overhang if it occurs. Calculate the value of, a , = 10% of least horizontal dimension or $0.4 \cdot h$, whichever is smaller, but not less than either 4% of least horizontal dimension or 3 feet. If your roof height is less than 30 feet, but not exactly 15, 20, or 25 feet, you will need to go to the next higher roof height. If your Mean Roof Height is higher than 30 feet, these charts do not apply. Review the diagram which illustrate the wall and roof zones and determine the wind zone in which the component is located. Determine the tributary area of the component. If the tributary area falls in between values, use the value of the smaller tributary area. Select the positive and negative wind pressures corresponding to the wall or roof zone where your component is located. Door pressures shown are for the most common door sizes and are worst case for heights ≤ 30 Feet.

Wall Pressure For All Roof Types												Garage/Door Pressures				
Mean Roof Height	15 Ft						20 Ft						<= 30 Ft			
Tributary Area	10	20	35	50	100	500	10	20	35	50	100	500	Effective Wind Area		Positive	Negative
Wall Positive Pressure	38.1	36.3	35.0	34.1	32.4	28.4	40.4	38.5	37.1	36.1	34.3	30.1	Width	Height		
Zone 4 Negative Pressure	-41.4	-39.6	-38.2	-37.3	-35.6	-31.6	-43.8	-42.0	-40.5	-39.6	-37.7	-33.5	8	8	38.6	-48.2
Zone 5 Negative Pressure	-51.0	-47.5	-44.8	-43.0	-39.6	-31.6	-54.0	-50.4	-47.5	-45.6	-42.0	-33.5	10	10	37.4	-45.7
Mean Roof Height	25 Ft						30 Ft						14	14	35.4	-41.8
Tributary Area	10	20	35	50	100	500	10	20	35	50	100	500	9	7	38.7	-48.3
Wall Positive Pressure	42.2	40.3	38.8	37.8	35.9	31.5	43.9	41.9	40.3	39.3	37.3	32.8	16	7	37.0	-45.0
Zone 4 Negative Pressure	-45.8	-43.9	-42.4	-41.4	-39.5	-35.1	-47.6	-45.7	-44.1	-43.1	-41.1	-36.5	3	7	41.8	-54.6
Zone 5 Negative Pressure	-56.6	-52.8	-49.7	-47.8	-43.9	-35.1	-58.8	-54.7	-51.7	-49.6	-45.7	-36.5	6	7	39.8	-50.6